

Quasi-periodic eruptions and repeated nuclear transients share one tight timescale relation across black hole masses.

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Background

Quasi-periodic eruptions and recurrent nuclear transients are repeating high-amplitude flares from galactic nuclei. Their recurrence times span hours to years, but the physical clock driving them remains unclear.

Recurrence time scatters widely with black hole mass, so what actually sets the flare cadence?

What this paper is selling

Compiling QPEs and RNTs, the authors find recurrence time scales almost linearly with flare rise time. This single relation unifies systems separated by orders of magnitude in timescale.

Key results to remember

- Recurrence time versus black hole mass is positive but heavily scattered, roughly 0.95 dex.
- Recurrence time scales nearly linearly with rise time, slope 1.00 ± 0.07 .
- The relation spans hours for QPEs to years for nuclear transients.

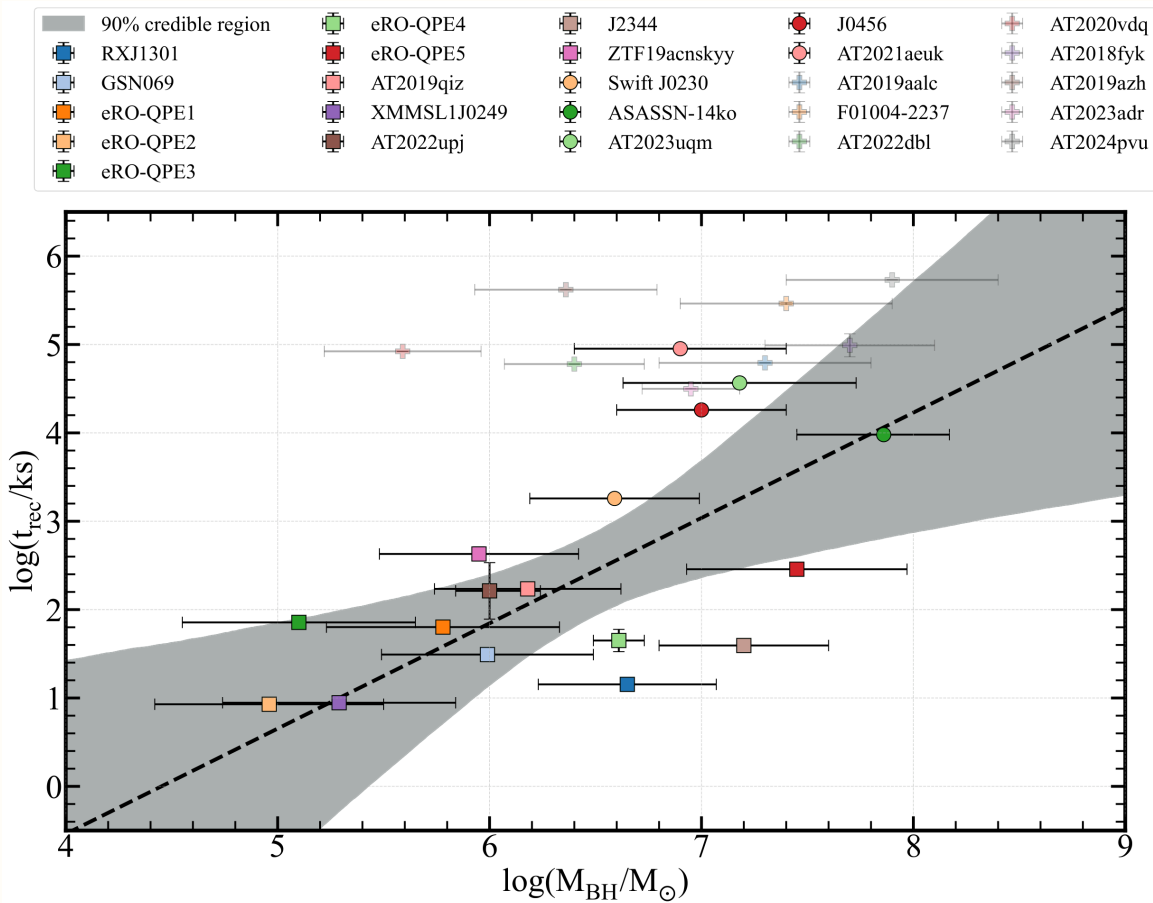


Figure 1 | Black hole mass versus recurrence time shows a positive but highly scattered trend across QPEs, RNTs, and rpTDE candidates.

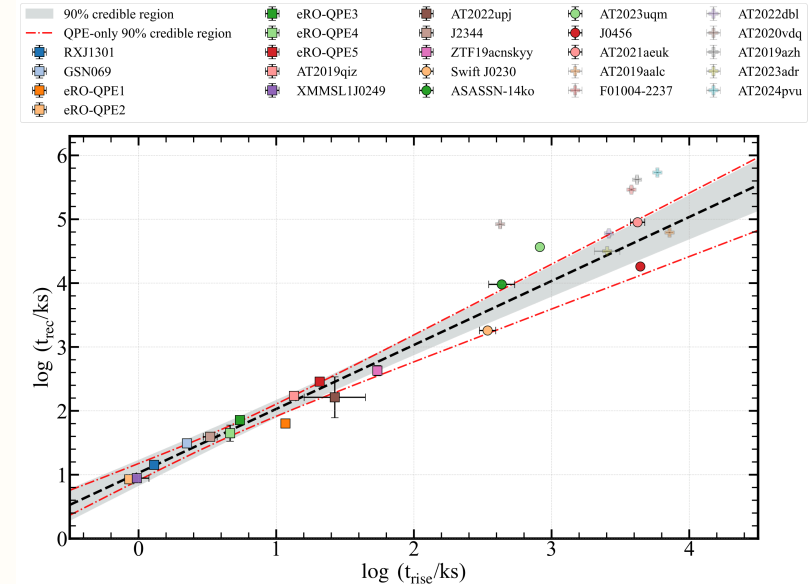


Figure 2 | Rise time versus recurrence time reveals a remarkably tight, nearly linear relation unifying QPEs and RNTs.